When Finite Differences Flip the Sign of Second-Order Generative Geometry: A Non-Recoverability Result and an Exact Instrument

Anonymous authors
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Abstract

Measuring the curvature of a generative model's image manifold by finite differences can return the wrong sign. On Stable Diffusion, the rank correlation between a Jacobian singular value and a second-order curvature ratio is -0.73 under exact forward-mode automatic differentiation and a sign-unstable -0.18 under a finite-difference estimate of the identical quantity. The same finite differences recover the first-order Jacobian accurately. The cause is an asymmetry. A central finite difference is well conditioned for a first derivative but *non-recoverable* for a second: truncation error and floating-point cancellation cannot be made small at the same step, leaving a relative-error floor of order $\sqrt{\varepsilon_{\mathrm{mach}}}$ that single precision makes large. On a StyleGAN generator the naive second-difference curvature misses the exact value by at least 45% at every fixed step, in fp32 and fp64 alike, while the first-order difference of the same code agrees with exact AD to 2.4%. We give the exact alternative, composed forward-mode AD, which returns the k-th-order pushforward in k forward passes at the memory of one. A survey of how the latent-geometry literature computes second-order quantities finds the dangerous $/\varepsilon^2$ estimate to be the obvious choice, yet one that most published methods sidestep by construction. We do not claim any published result is wrong. We characterize when finite differences fail for second-order generative geometry, give the first/second-order asymmetry as practitioner guidance, and supply a drop-in exact replacement.

1 Introduction

Let $G : \mathbb{R}^d \to \mathbb{R}^{3HW}$ be a trained generator. Treating the image set $\{G(z)\}$ as a manifold pulled back through G, a line of work studies its geometry: the pullback metric $M_z = J_z^\top J_z$ and its singular vectors (the local latent basis), geodesics between latent codes, and the curvature of the generated manifold (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018; Park et al., 2023; Saito & Matsubara, 2025). The first-order objects J_z and M_z are routine to compute. The second-order objects are not: the curvature, the second-order pushforward $\partial^2 G/\partial\alpha^2$ along a path, and the metric derivative $\partial_z M_z$ that appears in the geodesic ODE. These are where the numerical method can silently flip a result.

We start with a measurement (Section 2). A standard second-order quantity on Stable Diffusion returns the opposite sign depending only on whether the second derivative is taken by finite differences or by exact forward-mode AD. The exact method finds a clean, seed-consistent relationship. Finite differences, applied to the same quantity, are sign-unstable and wash it out. A practitioner who reached for the natural finite-difference estimate would draw the wrong qualitative conclusion.

The explanation is an asymmetry (Section 3). A central finite difference is well conditioned for a first derivative but non-recoverable for a second, because for the second derivative truncation and floating-point cancellation trade off and cannot be small together. The resulting relative-error floor is of order $\sqrt{\varepsilon_{\mathrm{mach}}}$. In the single precision that GANs and diffusion U-Nets run in, that floor is large, and on a strongly nonlinear generator it is larger still: at least 45% on StyleGAN, in both single and double precision (Section 4).

We contribute the reproducible sign-inversion phenomenon and the first/second-order asymmetry that governs it; an exact, constant-memory instrument for the higher-order pushforward, built by composing forward-mode AD; and a survey that locates which parts of the literature are exposed and which are safe. The contribution is a precise account of a real numerical hazard and the fix, not a claim that any published method is wrong; the survey finds that most route around the regime by construction (Section 7).

2 The sign-inversion phenomenon

On a Stable-Diffusion h-space pipeline we take a base latent, form the first-order Jacobian J of the (truncated) denoising map along K random h-space probes, and along its top-k right-singular directions u_i measure a second-order curvature ratio

$$\rho_i \;=\; \left.\frac{|\,\partial_\alpha^2 G(z+\alpha u_i)\,|}{|\,\partial_\alpha G(z+\alpha u_i)\,|}\right|_{\alpha=0},$$

the magnitude of the second-order pushforward relative to the first. We then ask a standard question: does curvature co-vary with the first-order singular value σ_i? We compute Spearman(σ,ρ) with the second derivative in ρ taken two ways. The first is exact: a composed JVP (nesting two forward-mode passes, defined in Section 5). The second is the finite difference of the JVP that a practitioner would naturally write.

Over 6 seeds, with the top 12 directions each, the two methods disagree in sign (Table 1). The exact instrument gives a strong negative correlation on five of six seeds, clustered in $[-0.94,-0.82]$, with seed 4 a near-zero outlier at -0.03; the mean is -0.73. Run through finite differences, the same ratio is sign-unstable across seeds and averages -0.18. This is not added noise but a destroyed relationship. The exact estimate is more negative than the finite-difference estimate on all six seeds (Wilcoxon signed-rank $p = 0.031$, the smallest value attainable at n=6), and the finite-difference estimate is sign-unstable at every step in the grid $\{0.1, 0.05, 0.01, 0.001\}$, with means between -0.08 and -0.34. No step reproduces the clean -0.73. A study built on the finite-difference route would report “no clear relationship,” the opposite of what the exact second-order estimate finds.

seed	0	1	2	3	4	5	mean
exact composed JVP	−0.85	−0.83	−0.92	−0.94	−0.03	−0.84	−0.73
FD-of-JVP(ε=0.05)	+0.18	−0.55	−0.78	−0.40	+0.55	−0.07	−0.18

Table 1: Spearman(σ,ρ) on Stable Diffusion, per seed. The exact second-order estimate is consistently negative; the finite-difference estimate of the identical quantity changes sign across seeds and averages near zero. We define ρ to expose the method dependence; the point is that the two numerical routes disagree on its sign, not the specific value of any prior work.

3 Why finite differences are non-recoverable for a second derivative

For a smooth scalar f the central second difference is $D_\varepsilon^2 f = \big(f(\alpha+\varepsilon) - 2f(\alpha) + f(\alpha-\varepsilon)\big)/\varepsilon^2$, and

$$D_\varepsilon^2 f - f''(\alpha) = \underbrace{\tfrac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4)}_{\text{truncation}} + \underbrace{O\big(\varepsilon_{\text{mach}}\,|f|/\varepsilon^2\big)}_{\text{cancellation}}.$$

The two terms cannot be small together. Minimizing their sum gives an optimal step $\varepsilon^\star \sim \varepsilon_{\text{mach}}^{1/4}$ and, under mild regularity ($f^{(4)}$ bounded, $f'' \neq 0$, $|f|$ of order one), a relative-error floor $2\sqrt{|f^{(4)} f|}/|f''| \cdot \varepsilon_{\text{mach}}^{1/2}$. The exponent $\frac{1}{2}$ is intrinsic to the operation; the constant carries the generator’s nonlinearity $|f^{(4)} f|/f''^2$, which is large for a deep network. In double precision the floor is about 10^{-8} in the benign (constant near one) case. In the single precision generators use ($\varepsilon_{\text{mach}} \approx 1.2\times10^{-7}$ for fp32) it is about 3×10^{-4} benign, and far worse for a strongly nonlinear G, where Section 4 measures $\geq 45\%$. The floor is a property of the operation in fixed precision, not of any architecture, so the hazard is general; we demonstrate its magnitude on two generator families.

The first derivative behaves differently. The central first difference $\big(f(\alpha+\varepsilon)-f(\alpha-\varepsilon)\big)/2\varepsilon$ has truncation $O(\varepsilon^2)$ and cancellation $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, a much milder floor $\varepsilon_{\text{mach}}^{2/3}$ with a wide usable window. First-order Jacobians are therefore safe to finite-difference while second-order curvature is not. This is also the control we use throughout: wherever the first-order finite difference agrees with the exact JVP, the pipeline is correct, and any second-order disagreement is real rather than a bug.

To check the constants against ground truth, let $f(\alpha)=[\,e^{0.7\alpha},\sin 3\alpha,(\alpha+0.4)^3,\tanh 2\alpha,\alpha^2\cos\alpha\,]$, whose derivatives are known in closed form. At $\alpha=0.3$ in double precision (Table 2) the central second-order relative error is U-shaped: it bottoms at 8.6×10^{-9} near $\varepsilon=10^{-4}$ and diverges to 0.29 by $\varepsilon=10^{-8}$. The composed JVP reaches 2.5×10^{-17}, so the best finite difference is 3.5×10^{8} times worse.

ε	central FD 2nd-order rel. err.	
10^{-1}	7.9×10^{-3}	(truncation)
10^{-2}	8.0×10^{-5}	
10^{-4}	8.6×10^{-9}	(best)
10^{-6}	3.0×10^{-5}	
10^{-8}	2.9×10^{-1}	(cancellation)
composed JVP	2.5×10^{-17}	(exact)

Table 2: Second-order accuracy on the analytic toy at $\alpha=0.3$, double precision. Finite differences never approach the exact composed JVP.

4 Quantification on a real generator

On a real generator the failure is worse than the toy predicts, and for an extra reason. We compare the naive second-difference curvature a practitioner would write, $D_\delta^2 G=\big(G(z{+}\delta u)-2G(z)+G(z{-}\delta u)\big)/\delta^2$, against the exact composed JVP $\partial_\alpha^2 G(z{+}\alpha u)$ on a StyleGAN-bedroom generator at 256×256, over 6 directions (3 annotated attributes, 3 random) and 16 samples, sweeping δ (Table 3). The exact curvature is a definite, step-free number, mean 1.17. Holding a single step across all samples, as a practitioner does, the finite-difference mean relative error never drops below 45.2% (best step $\delta=3$, bracketed on both sides); it is 267% at $\delta\approx0.3$ and explodes as $\delta\to0$. Allowing an oracle per-sample step the median best error is still about 9%, against the composed JVP's zero.

Precision does not rescue it. Rerunning the entire sweep in fp64 leaves the floor unchanged, 47.1% in both fp32 and fp64. The mechanism is not cancellation alone. At usable steps the error is truncation and is precision-blind. At tiny steps fp32 adds catastrophic cancellation, but fp64, which removes that, instead exposes the generator's non-smoothness: across a LeakyReLU kink the chord-scale second difference does not converge to the pointwise pushforward, and the fp64 error grows past 3000% as $\delta\to0$. So no step and no precision recovers the exact second-order pushforward, which the composed JVP returns directly.

The control confirms the instrument is correct. The first-order central difference of the same pipeline agrees with the exact first-order JVP to 2.4%, so the second-order failure is genuine.

5 The exact instrument

Forward-mode AD evaluates, for a map f and tangent v, the Jacobian–vector product $\partial f(x)[v]$ alongside $f(x)$ in one pass, without forming the Jacobian. With $f=G$, $x=\gamma(\alpha)$, and $v=u$ this gives the exact first-order pushforward. The second-order term is the directional derivative of the first, so composing forward mode with itself,

$$\partial_\alpha^2 G(\gamma(\alpha))=\partial_\alpha\big[\partial G(\gamma(\alpha))[u]\big][u],$$

is exact and is evaluated by nesting two JVP calls: in PyTorch, `torch.func.jvp` of a function that itself returns a JVP. No Jacobian is stored, so memory stays at one forward pass, and the same nesting yields any order in a constant number of passes. Composing forward-mode passes for higher-order directional derivatives

step δ	$\lvert D_\delta^2 G \rvert$	2nd-order rel. err.	1st-order rel. err.
8	7.1×10^{-3}	74.5%	63.8%
3	2.7×10^{-2}	45.2% (floor)	44.1%
1	9.9×10^{-2}	106.4%	26.2%
0.3	3.7×10^{-1}	266.9%	12.5%
0.1	1.06	622.8%	5.5%
0.02	4.05	6.8×10^{3}%	2.4% (1st-order min)
10^{-3}	4.09×10^{2}	2.0×10^{6}%	139%
composed JVP	1.17	exact (step-free)	—

Table 3: StyleGAN-bedroom, fp32. Columns are means over 6 directions $\times 16$ samples at each fixed step. The relative-error column is the mean per-direction relative error; $\lvert D_\delta^2 G \rvert$ is the aggregate magnitude, shown for scale and not the quantity the relative error is computed from. The naive second difference never gets within 45% of the exact curvature at any fixed step, while the first-order difference of the same code agrees to 2.4%. Rerunning in fp64 leaves the best-step floor unchanged (47.1% fp32 vs. 47.1% fp64): the floor is truncation at usable steps and generator non-smoothness at tiny steps, not a single-precision artifact.

is standard algorithmic differentiation (Griewank & Walther, 2008); the contribution here is to apply it as the exact ground truth against which the finite-difference practice fails. The instrument is generator-agnostic: we ran it unmodified on StyleGAN1 and 2 and on a latent-diffusion U-Net. Unlike reverse-mode Hessian materialization, whose memory is $O(3HW)$, its cost is independent of output dimension.

6 Related work and survey

The finite-difference step-size tradeoff is classical numerical analysis (Press et al., 2007), and higher-order forward-mode AD is standard (Griewank & Walther, 2008); what has been missing is an account of where the tradeoff bites in the differential geometry of generative models, and how often. Table 4 classifies representative latent-geometry papers by how they obtain their second-order or geodesic quantities. The explicit $/\varepsilon^2$ second-difference, the dangerous regime of Section 3, is the obvious estimate for a practitioner adding a curvature measurement, but most published methods never form it. They keep the metric first order, replace the geodesic ODE with a discretized energy whose exact gradient is a well-conditioned discrete Laplacian rather than a $/\varepsilon^2$ curvature, use analytic Christoffel symbols, or differentiate the energy with autodiff (Yang et al., 2018); the first-order constructions (Park et al., 2023; Chen et al., 2018; Kwon et al., 2023) never take a second derivative. The one exposed published practice is the first-order finite difference of a network derivative, such as a score-Jacobian–vector product, which sits on the well-conditioned $\varepsilon_{\text{mach}}^{2/3}$ side of Section 3.

The field is mostly safe by construction, not by caution, and that safety is a property of the current formulations rather than a law. As curvature- and geodesic-level analysis of diffusion models grows, for instance the energy-and-score geometry of Saito & Matsubara (2025) or any second-order extension of the first-order bases of Park et al. (2023), the $/\varepsilon^2$ estimate becomes the obvious next step. We intend Table 4, the asymmetry, and the drop-in instrument as guardrails for that emerging work.

Guidance. Finite-difference first-order quantities freely: choose a step near $\varepsilon_{\text{mach}}^{1/3}$ and expect a few digits. Never finite-difference a second derivative. In single precision no step gives even one reliable digit on a deep generator (Section 4), and the error can flip the sign of a downstream conclusion (Section 2). Use composed forward-mode AD: it is exact, costs one extra forward pass per order, and adds no memory. It also gives a free check. If the first-order finite difference and the JVP disagree, the pipeline is wrong; if they agree, any second-order disagreement is real.

Paper	2nd-order quantity	how obtained
Shao et al. (2018)	discretized geodesic energy	exact discrete-energy gradient (Laplacian), coarse T; no $/\varepsilon^2$
Saito & Matsubara (2025)	geodesic (energy)	first-order score-differences $\|s(x_{i+1})-s(x_i)\|^2$; no $/\varepsilon^2$
Arvanitidis et al. (2018)	geodesic ODE (Christoffel)	analytic ∂M, numeric ODE integ.
Yang et al. (2018)	geodesic energy	autodiff
Park et al. (2023)	none (first-order basis)	autodiff power iteration
Chen et al. (2018)	none (first-order metric)	—
Kwon et al. (2023)	none (first-order)	CLIP-grad + MLP

Table 4: How representative latent-geometry papers obtain second-order quantities. The explicit $/\varepsilon^2$ second difference (Sections 3, 4) is largely avoided in published work, but it is the obvious choice for a practitioner adding a curvature measurement, which is where the hazard bites.

7 Scope and limitations

One limitation matters more than the rest. We show that the measurement can flip, not that any published number is wrong; the survey's point is that the field mostly avoids the trap. The sign inversion of Section 2 is on a ratio ρ that we define to expose method dependence, not a quantity from prior work. The remaining caveats are routine. The non-recoverability result is specific to second-order finite differences; first-order finite differences, including score-JVPs, are well conditioned, as we state. The instrument is a thin exact wrapper around forward-mode AD, with no algorithmic novelty; its value is in making exact higher-order pushforward routine and in the account of when the finite-difference alternative fails. Finally, we show no downstream task win: this is a measurement-correctness result, and whether accurate second-order geometry changes downstream conclusions is left to future work.

8 Conclusion

A second derivative of a deep generator cannot be recovered by finite differences: no step and, on a real generator, no precision closes the gap to the exact value, and the error is large enough to flip the sign of a measured geometric relationship. A first derivative is safe, which gives both a clean diagnostic and a reason the failure is easy to miss. The fix is a one-line composition of forward-mode AD that returns the exact higher-order pushforward at the memory of a single forward pass. Most current latent-geometry methods avoid the dangerous estimate by construction, but as second-order and geodesic analysis of diffusion models grows, the exact instrument is the principled default.

Reproducibility statement

We release the exact-instrument module, the analytic referee, and the StyleGAN and Stable-Diffusion scripts, with fixed seeds and the emitted `metrics.json` behind every table. The generators are the public StyleGAN-bedroom-256 weights and Stable Diffusion v1.5; all experiments run on a single 8 GB GPU. The code link is withheld for anonymous review and will be added for the camera-ready version.

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